Advisor briefing: progress since October 2

Progress since October 2 and the manuscript direction

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Main message

We have completed the full-range Single gene-count sensitivity study and requested CpG variability inventory. Prediction stays broadly similar as genes increase; programs are particularly stable from about 5k to 10k genes. Larger CpG fits exposed a numerical failure that is repaired experimentally, but reliable convergence remains the main methodological obstacle.

The benchmark is retained: Single external C=0.627 across five cohorts with $\alpha_F = 0$ (survival does not guide its loading updates). Joint ICGC primary C=0.685 versus expression-only C=0.676; both fits are unconverged and incremental predictive benefit is unproven.

What increasing the feature count shows

Same 273 TCGA+CPTAC training patients, $K = 7$, no pruning and cold starts; all 24 full-training/CV gene fits converge.

Shared genes	Conditional CV C ± SE	Retained programs
2,064	0.608 ± 0.046	4
5,001	0.577 ± 0.027	4
10,000	0.609 ± 0.040	4
17,900 (all)	0.607 ± 0.019	3

There is no clear predictive gain. Screening/scaling preceded folds; this is conditional CV, not fully nested preprocessing evaluation. No count is adopted.

Expansion	Matched programs	Common-gene loading $\|r\|$	Risk r
2,064 → 5,001	4	0.115–0.962	0.655
5,001 → 10,000	4	**0.960–0.994**	**0.972**
10,000 → 17,900	3	0.737–0.915	0.667

Strong agreement over 5k–10k does not mean every program is stable across the entire range. Constant/unmatched factors have undefined correlations.

The CpG inventory covers 305,492 raw probes; 305,352 are model-eligible. Observed mean/SD histograms and both requested filter-overlap summaries are complete; no filter or SD cutoff is adopted. The 10k methylation-only fit caps at 500 with zero credible coefficients. Original 25k fitting fails from Cox risk-set underflow. Stable log-space arithmetic removes that error, but the same case still caps at 500 with extreme predictors: training C=0.984 is not a validated improvement. Larger counts and matched CV/joint curves remain unfinished.

What this means for the method and manuscript

Survival-guided loading updates are implemented experimentally, but a fully validated supervised replacement, rank selection and added multimodal value remain unresolved. The next method work addresses prior/objective consistency; software tests alone do not establish a successful scientific remedy.

The full write-up describes the method, benchmark results and provisional biology. Existing evidence supports expression-defined programs and methylation-dominant structure, not its biological cause or added prognostic value. Final labels need annotation/enrichment and composition/batch checks.

Advisor discussion: agree on the manuscript's central contribution and prioritize stable supervision before the remaining matched feature curves. A computational-methods paper with a convincing PDAC evaluation could fit *Bioinformatics Advances*; a stronger statistical-methods emphasis could fit *Statistics in Biosciences*. These are provisional scope-based options, not a submission decision. The current write-up can support drafting now; scientific validation still determines the claims.

Accompanying documents provide the detailed evidence report, the full method/results/biology write-up, and a journal shortlist with sources and remaining manuscript-readiness requirements.

Guide to the project write-up

Start with the structured summary, then use four questions to align on the current evidence and the intended manuscript:

Read	Main question for discussion
Methods and model-comparison table	What do activities, loadings and projection scores mean, and which formulation supports the reported results?
External validation and feature-count results	How convincing are prediction and program stability relative to the actual comparator?
Biological findings	Do basal-like/classical versus stromal/immune programs give useful biological separation, and what would establish added methylation value?
Discussion and continuing work	What is the strongest defensible manuscript claim now, and what evidence is essential before submission?

Proposed narrative: Bayesian program models connect cross-cohort risk prediction with interpretable tumor and microenvironmental structure, and provide a framework for testing what broader molecular inputs contribute. The present evidence supports that starting point. The stronger claim that survival improves program discovery requires the ongoing methodological validation; neither uncertainty calibration nor added multimodal value is established merely by the model's formulation.